Blockchains & Cryptocurrencies

Anonymity

Instructor: Matthew Green
Fall 2026

Logistics

- Project draft 1is due March 30, 11:59 pm
- Assignment 2 part 2 is coming out today, due April 4
- Midterm is April 8 in class
  - Will cover everything through the previous class: Bitcoin, Ethereum, Privacy; weds we will go over topics

News?

News?

The Motley Fool

XRP and Ethereum Are Both Pivoting to Privacy. Is That a Reason to Buy Either?

Alex Carchidi, The Motley Fool
February 28, 2026 • 4 min read

XRP-USD +3.88% | INTC +0.98% | NVDA +2.06% | ETH-USD +4.29%

When you buy something with cash, you can usually assume, at least in theory, that the transaction is mostly private. In crypto, the opposite is true. Blockchains are, in a sense, public databases, which means that every time you buy, sell, or transfer crypto, it's typically quite easy for anyone to see which wallet address sent what to whom.

That's not exactly an ideal state of affairs, which is why many leading

Anonymity

Anonymity in computer science

Anonymity = pseudonymity + unlinkability

Different interactions of the same user with the system should not be linkable to each other

Pseudonymity vs anonymity in forums

Reddit: pick a long-term pseudonym

vs.

4Chan: make posts with no attribution at all

Why is unlinkability needed?

1. Many Bitcoin services require real identity

1. Linked profiles can be deanonymized by a variety of side channels

Defining unlinkability in Bitcoin

- Hard to link different addresses of the same user
- Hard to link different transactions of the same user
- Hard to link sender of a “payment” to its recipient

Quantifying anonymity

Anonymity set: Anonymity set of a transaction T is the set of transactions which an adversary cannot distinguish from T.

To calculate anonymity set:

- define adversary model
- reason carefully about: what the adversary knows, does not know, and cannot know

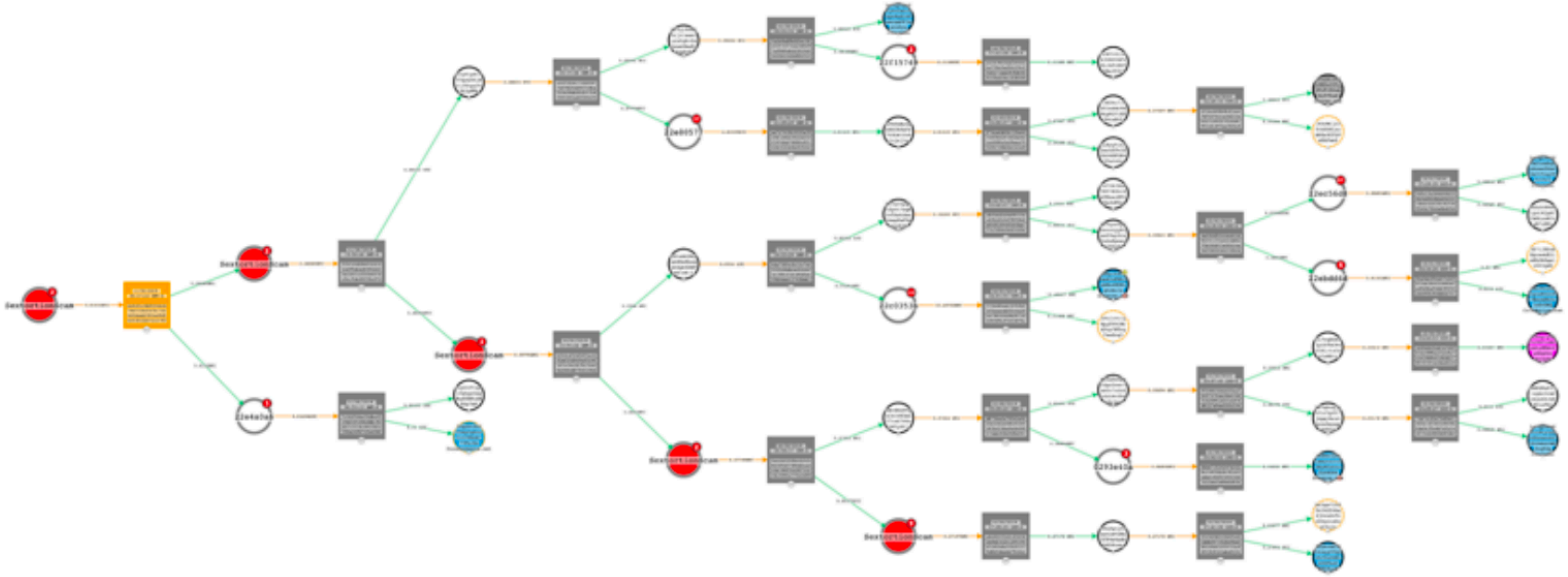
Why anonymous cryptocurrencies?

Block chain based currencies are totally, publicly, and permanently traceable

Without anonymity, privacy is <u>much worse</u> than traditional banking!

Whale Moves $1.16B Bitcoin in Largest-Ever Dollar Transaction

beincrypto.com | 1d



Additionally, 1PXf also sent 0.01359 BTC to GoURL, another 0.0098 BTC to Empire Market (Dark Market), and 0.00507 BTC to Bovada (Gambling). Plus, it sent 0.1405 BTC to 3PFFkzVgbxhwHejFyZeeyXKBFG7dL5gRcj, which still maintained that balance as of December 6.

Several recipients of the emails reported that their payment instructions cited the 1PXf address

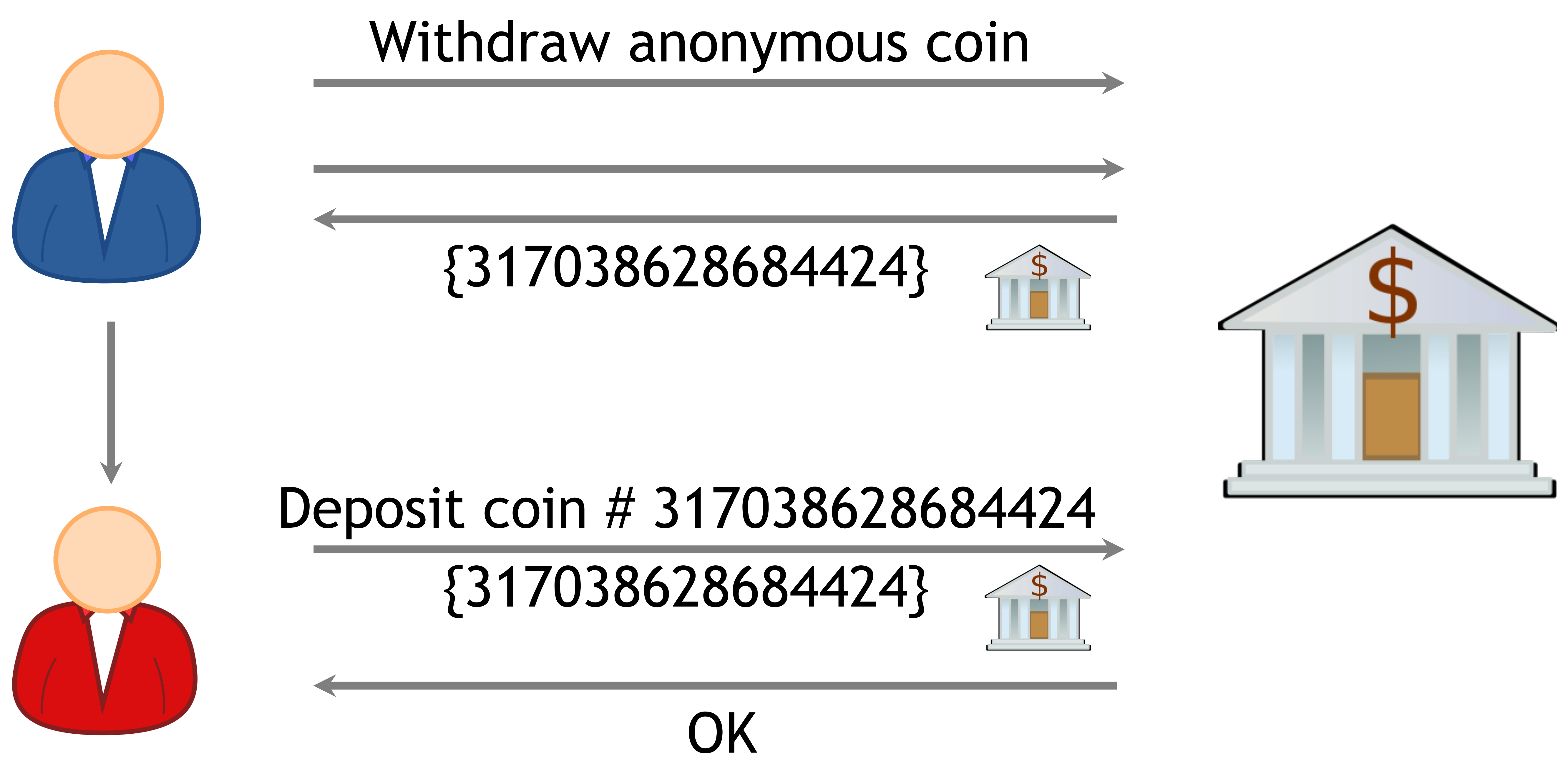
Anonymous e-cash: history

Introduced by David Chaum, 1982

<u>Blind signature</u>: a two-party protocol to create digital signature without signer learning which message is being signed

- An example of secure two-party computation

Anonymous e-cash via blind signatures



User	Balance
…	…
	9
…	…
	6

Spent coins
…
31703862…

Bank cannot link the two users

Anonymity & decentralization: in conflict

- Interactive cryptographic protocols with bank are hard to decentralize
  - Later: Zerocoin, Zerocash, Monero overcome this challenge by using non-interactive cryptographic techniques
- Decentralization often achieved via public traceability to enforce security

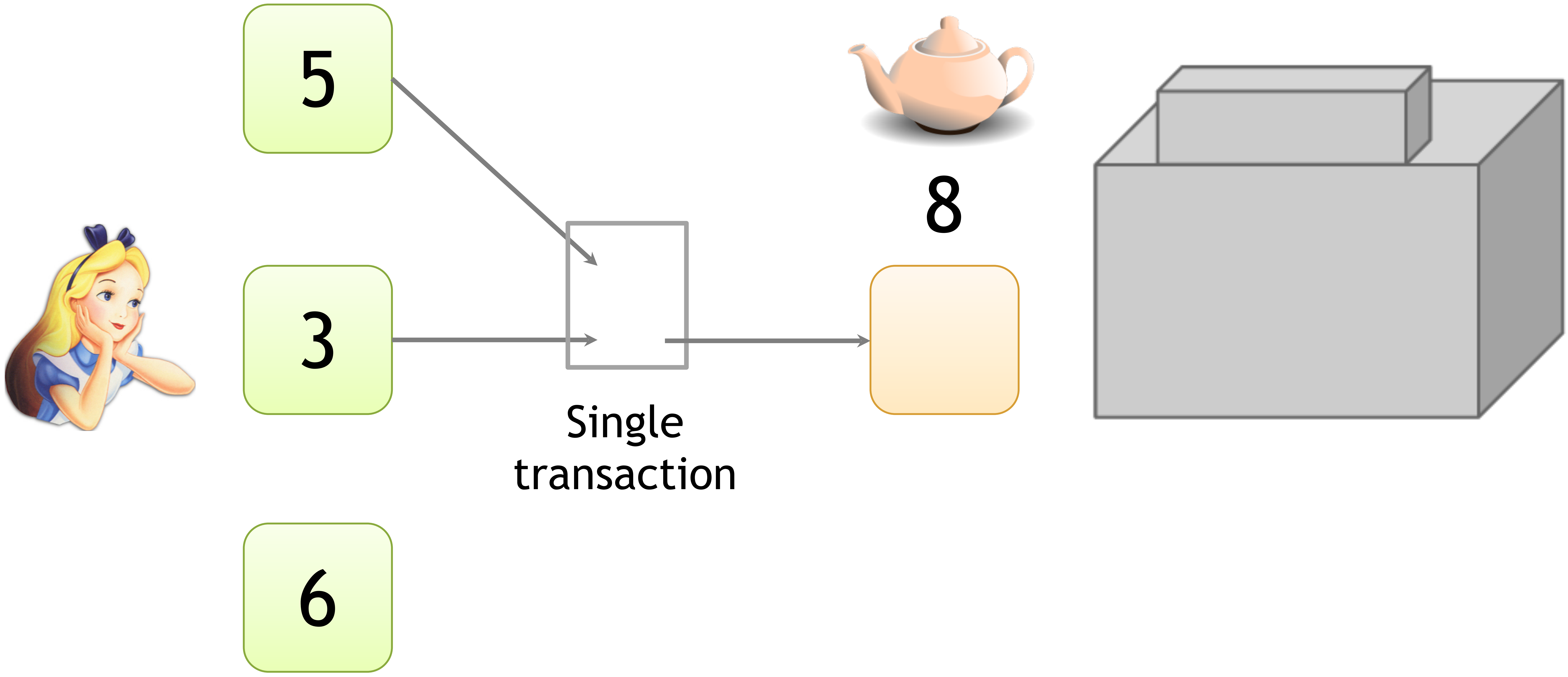
How to de-anonymize Bitcoin

Trivial to create new addresses in Bitcoin

Best practice: always receive at fresh address

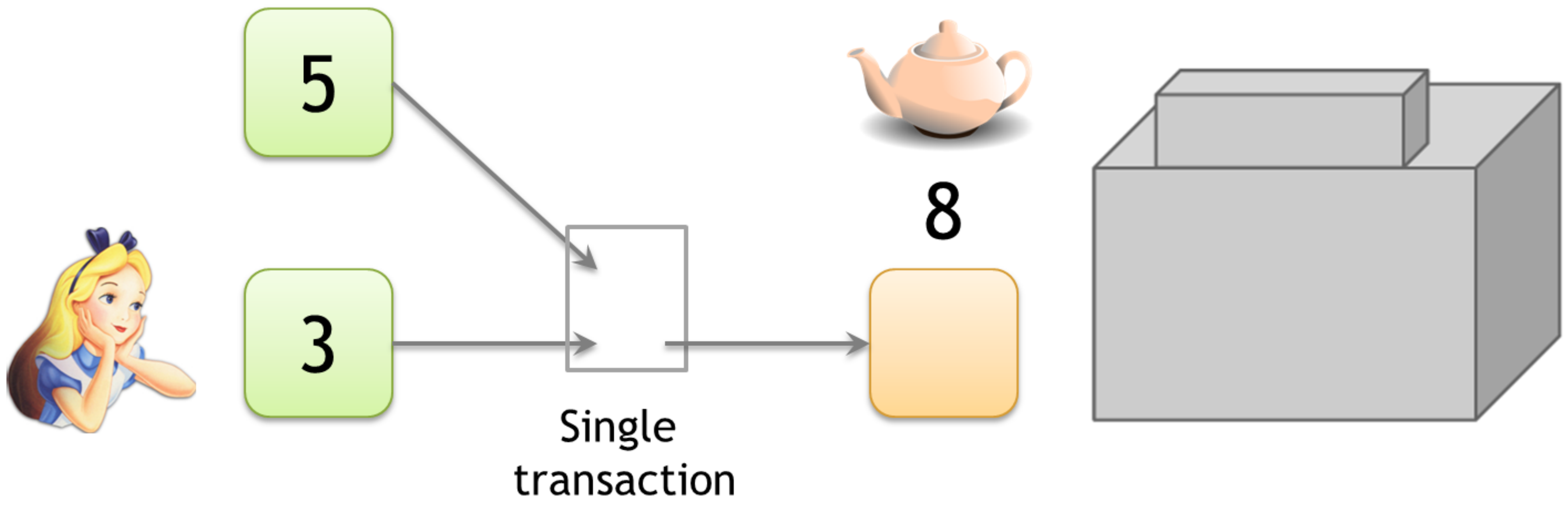
So, unlinkable?

Alice buys a teapot at Big box store



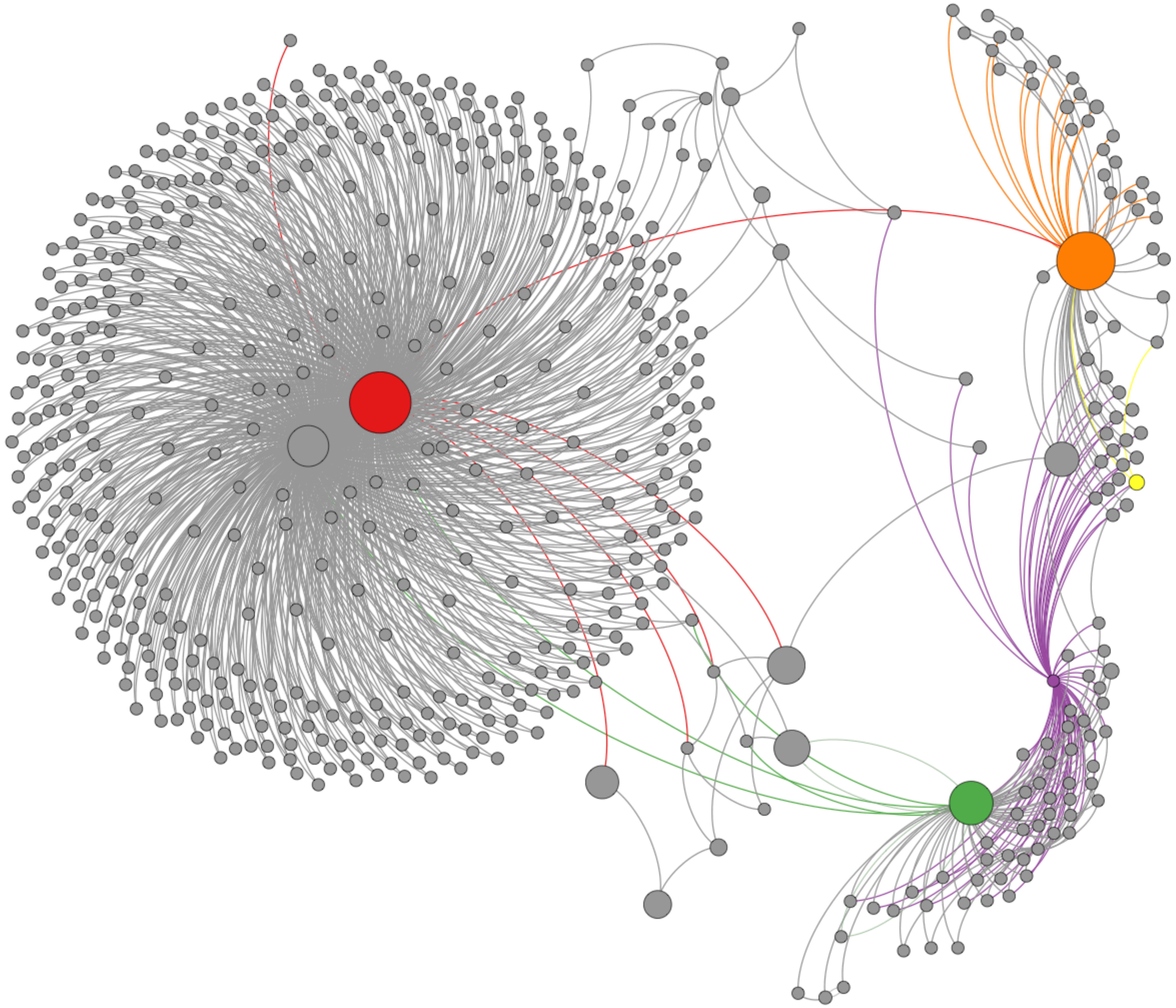
Linking addresses

Shared spending is evidence of joint control



Addresses can be linked transitively

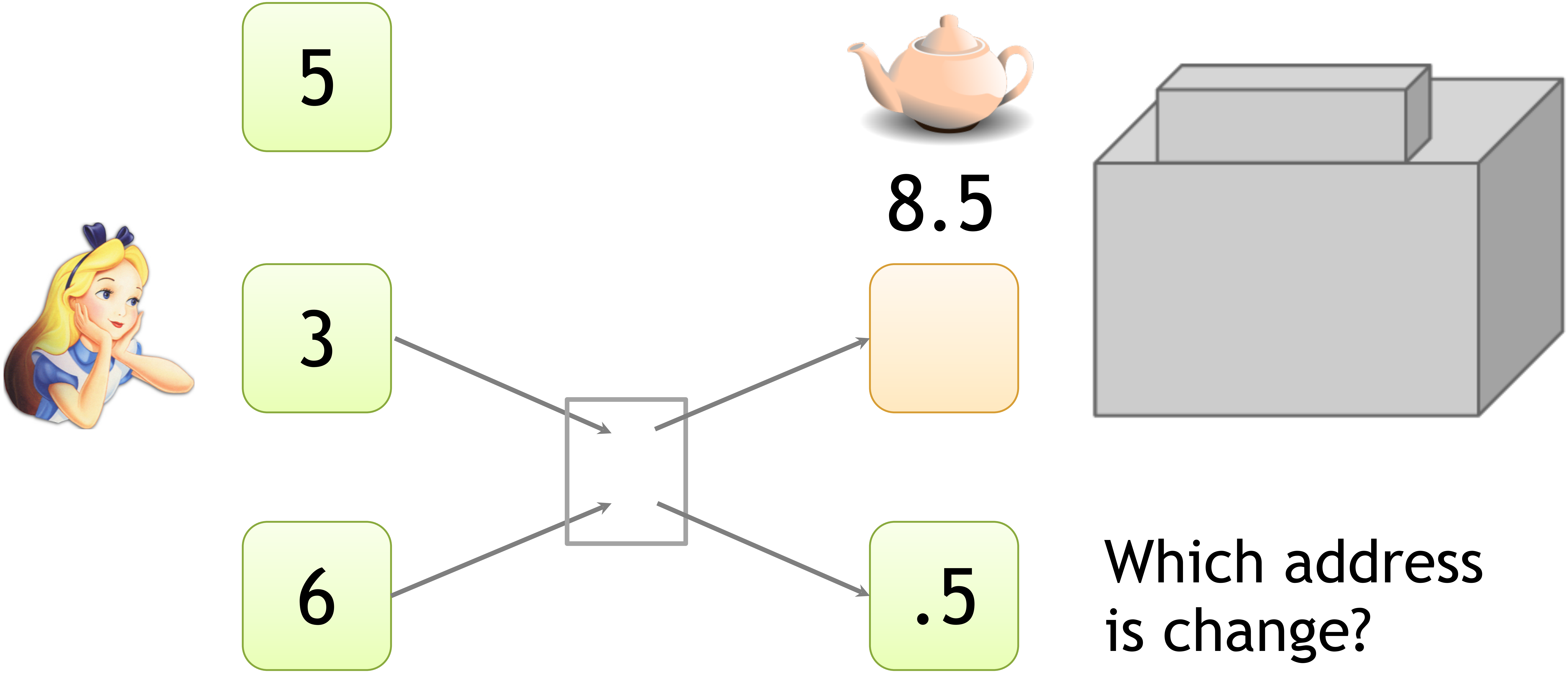
Clustering of addresses



An Analysis of Anonymity in the Bitcoin System

F. Reid and M. Harrigan
PASSAT 2011

Change addresses



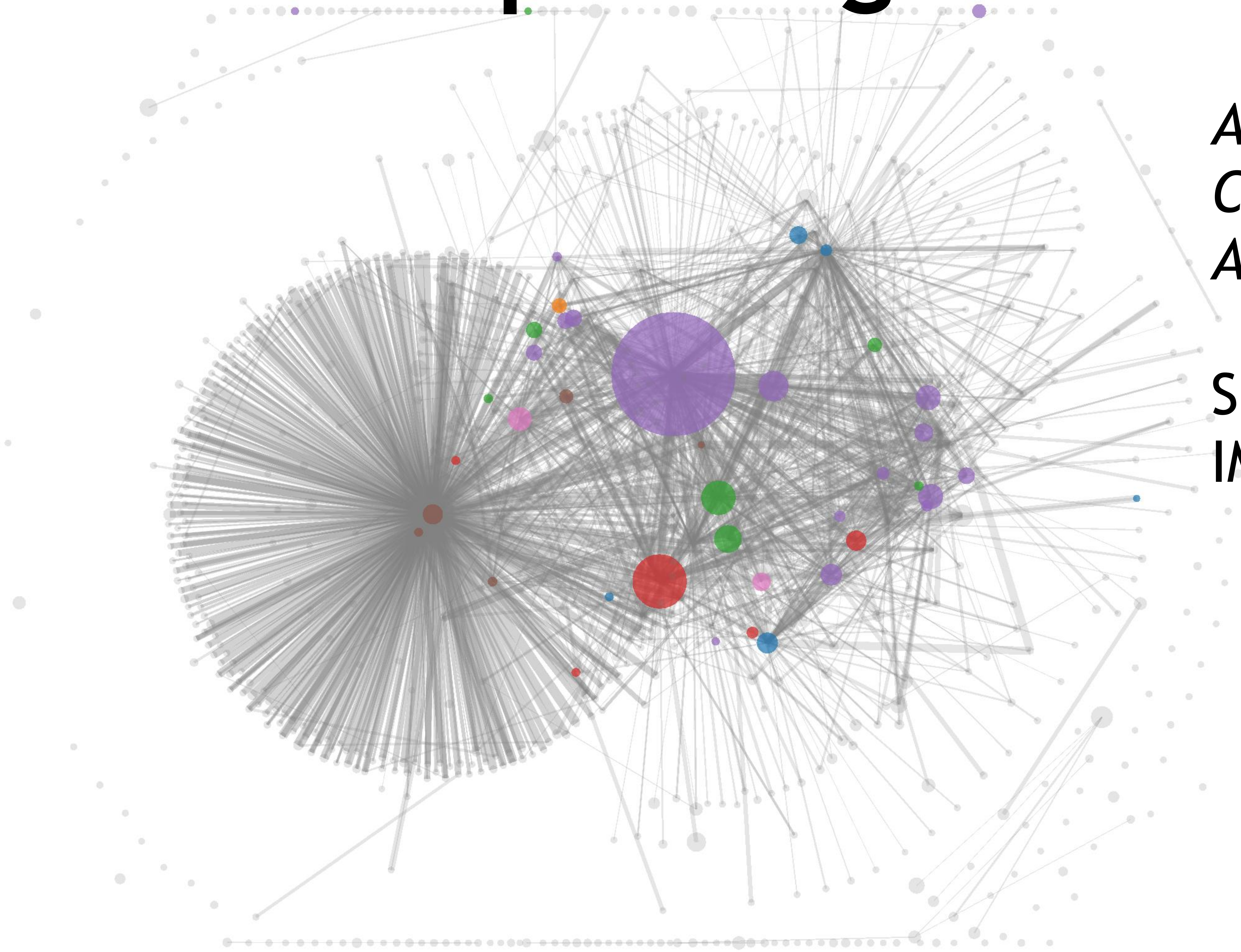
Which address is change?

“Idioms of use”

Idiosyncratic features of wallet software

e.g., each address used only once as change

Shared spending + idioms of use



A Fistful of Bitcoins: Characterizing Payments Among Men with No Names

S. Meiklejohn et al.
IMC 2013

To tag service providers: transact!

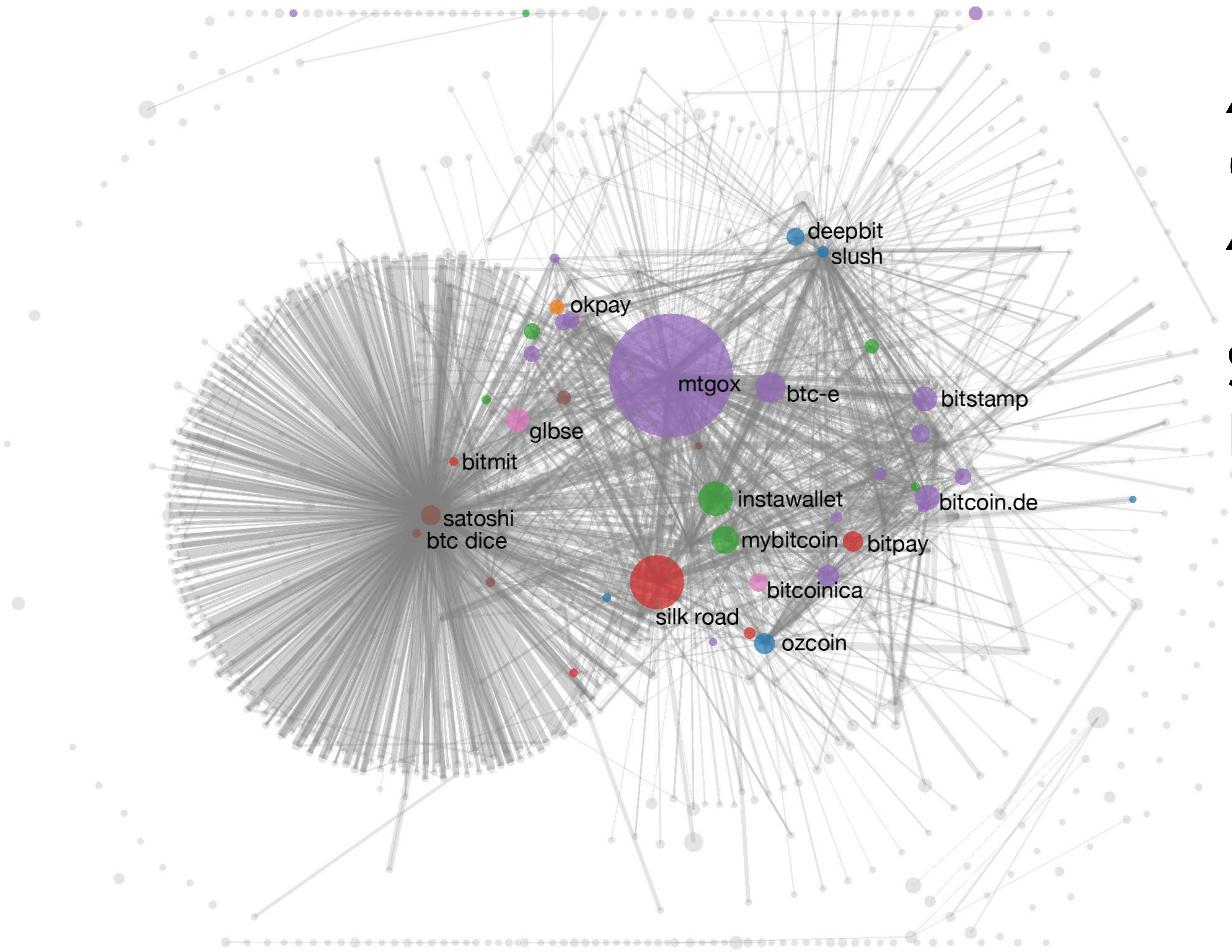
A Fistful of Bitcoins: Characterizing Payments Among Men with No Names

S. Meiklejohn et al.

344 transactions

- Mining pools
- Wallet services
- Exchanges
- Vendors
- Gambling sites

Shared spending + idioms of use



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From services to users

1. High centralization in service providers

   Most flows pass through one of these – in a traceable way

2. Address – identity links in forums

Achieving Anonymity

Approaches

- **Mixing**: Pool in multiple transactions (ideally same value), and then create new transactions
  - Centralized: E.g., online wallets
  - Decentralized: E.g., CoinJoin
  - Untrusted intermediary using crypto: Tumblebit
- **New cryptocurrencies:**
  - Using Zero-knowledge proofs: Zerocoin and Zerocash
  - Using Ring signatures: Monero

Approaches

- **Mixing**: Pool in multiple transactions (ideally same value), and then create new transactions
  - Centralized: E.g., online wallets
  - Decentralized: E.g., CoinJoin (e.g., implementation: Dash)
  - Untrusted intermediary using crypto: Tumblebit
- **New cryptocurrencies:**
  - <u>Using Zero-knowledge proofs: Zerocoin and Zerocash</u>
  - Using Ring signatures: Cryptonote (e.g., implementation: Monero)

Ingredient: Commitments

- Like a digital "envelope": allows you to commit to a message value, without revealing what it is
  - C = Commit(message; randomness)
  - **Hiding**: given a commitment, can't see what message it is, until I "open" the commitment and reveal it to you
  - **Binding**: giving you a commitment "binds" me to a specific message/value. I can't change my mind when I open it.

Hash commitments

- Example: hash commitments
  - Pick some random "salt" (e.g., 256 bits) **r**
  - Compute C = Hash(message || r)
  - To open: reveal (message, r), verifier checks hash

Hash commitments

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  - Pick some random "salt" (e.g., 256 bits) **r**
  - Compute C = Hash(message || r)
  - To open: reveal (message, r), verifier checks hash
  - **Hiding:** For many common hash functions, it is hard to "invert" C to get back (message, r) so this is hiding — *but what if we didn't add randomness?*
  - **Binding:** Finding another pair (message, r) that hashes to C would require finding a <u>collision</u> in the hash function

Pedersen Commitments

- We need a cyclic group. $G = \langle g \rangle$ where it is hard to find *x* given (g, g^x) — AKA the "discrete log problem" (DLP) is hard
  - E.g., G can be a subgroup of a finite field $\{1, \ldots, p-1\}$ where exponentiation/multiplication are modulo *p*
  - We also need two public generators: g, h *such that nobody knows the discrete log of g resp. h (vice versa)*
  - Commitment to message: pick random $r \in \{0, \ldots, groupOrder - 1\}$ compute: $C = g^m \cdot h^r$
  - To open the commitment, simply reveal (m, r)

Pedersen Commitments

- Why is this secure?

  - **Hiding:** If g, h are generators, then h^r is a random element of the group, so. C is too

    $C = g^m \cdot h^r$

Pedersen Commitments

- Why is this secure?

  - **Hiding:** If g, h are generators, then h^r is a random element of the group, so. $C = g^m \cdot h^r$ is too.

  - **Binding:** Let q be the group order. Let $h = g^x$ for some unknown *x*. Assume an attacker can find (m, r) != (m’, r’) such that $g^m h^r = g^{m'} h^{r'}$. Then it holds that:

$$g^m g^{xr} = g^{m'} g^{xr'}$$

and thus,

$$m + xr = m' + xr' \ mod \ q$$

We can solve for *x*, which means solving the DLP!

Confidential Transactions

- Pedersen commitments are <u>additively homomorphic:</u>
  - Commit to "m1": $C_1 = g^{m_1} h^{r_1}$
    Commit to "m2": $C_2 = g^{m_2} h^{r_2}$
  - Now multiply the two commitments together:

$$\begin{aligned} C_3 &= C_1 \cdot C_2 \\ &= g^{m_1} h^{r_1} \cdot g^{m_2} h^{r_2} \\ &= g^{m_1+m_2} h^{r_1+r_2} \end{aligned}$$

Notice that C3 is a commitment to the <u>sum</u> m1+m2
(under randomness r1+r2)

Confidential Transactions

- Introduced by Maxwell
  - Does not provide privacy for the identity of the input transactions, can be combined with CoinJoin or ringsides
  - Does allow you to hide the <u>value</u> of input transactions
  - Basic idea: use a Pedersen commitment to each transaction value, rather than revealing this in cleartext $C = g^v h^r$
  - Do a CoinJoin, and use additive property of Pedersen commitments to sum the values and then subtract each output commitment (board)

Zerocoin (MGGR14)

- Proposed as an extension to Bitcoin in 2014
  - Requires changes to the Bitcoin consensus protocol!
- I can take Bitcoin from my wallet
  - Turn them into 'Zerocoins'
  - Where they get 'mixed up' with many other users' coins
  - I can redeem them to a new fresh Wallet

Zerocoin

- Zerocoins are just numbers
  - Each is a digital commitment to a random serial number
  - Anyone can make one!

Minting Zerocoin

- Zerocoins are just numbers
  - Each is a digital commitment to a random serial number *SN*
  - Anyone can make one!

$$C = Commit(SN; r)$$

$$C = g^{SN} h^r \ mod \ p$$

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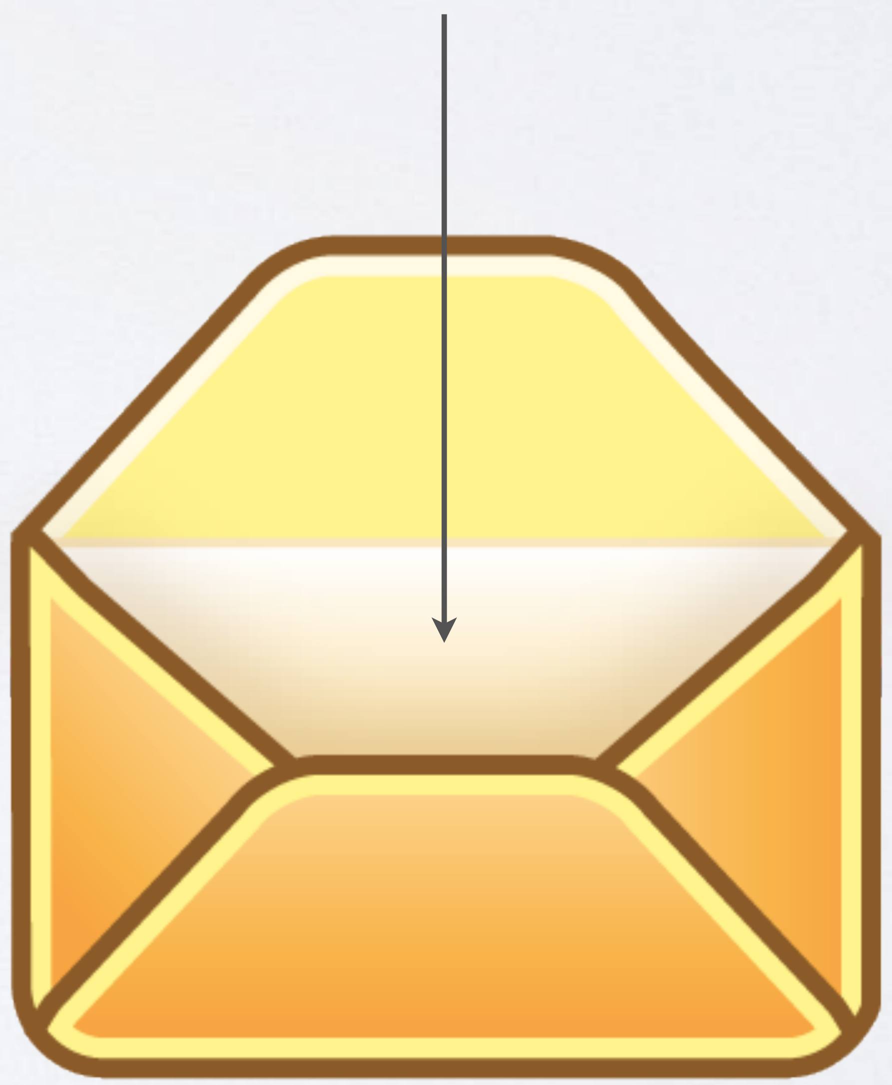
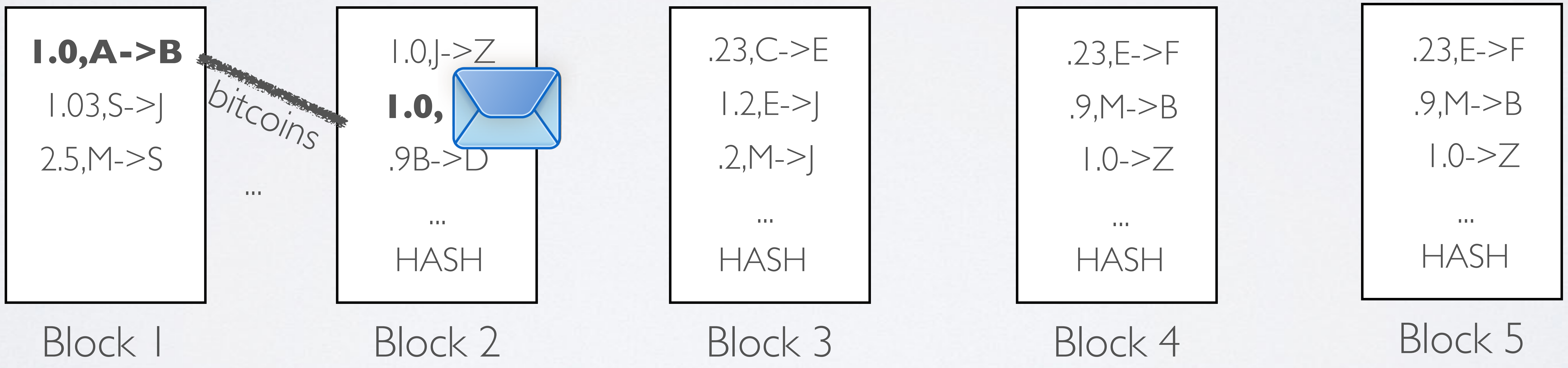


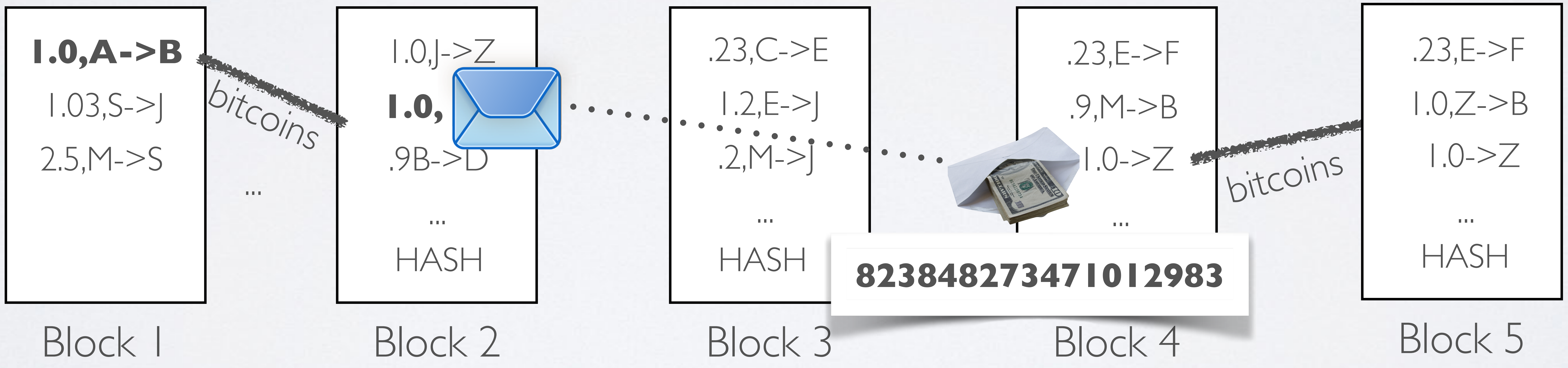
Minting Zerocoin

- Zerocoins are just numbers
  - They have value once you write them into a valid transaction on the blockchain
  - Valid: has inputs totaling some value e.g., 1 bitcoin



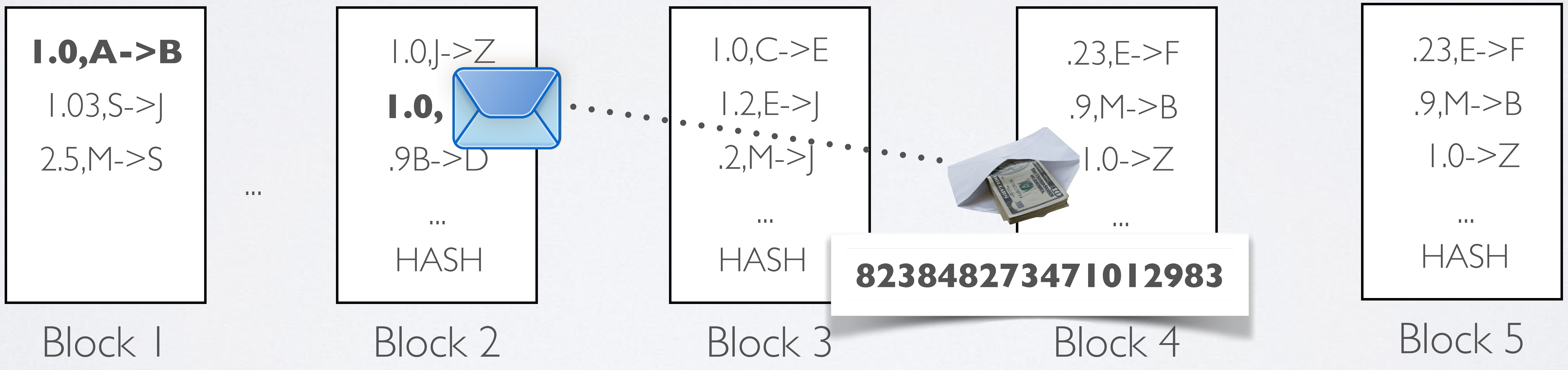
Redeeming Zerocoin

- You can redeem zerocoins back into bitcoins
  - Reveal the serial number &
    Prove that it corresponds to some Zerocoin on the chain
  - In exchange you get one bitcoin (if SN is not already used)



Spending Zerocoin

- Why is spending anonymous?
  - It's all in the way we 'prove' we have a Zerocoin
  - This is done using a <u>zero knowledge proof</u>



Spending Zerocoin

- Zero knowledge [Goldwasser, Micali 1980s, and beyond]
  - Prove a statement without revealing <u>any other information</u>
  - Here we prove that:
    (a) there exists a Zerocoin in the block chain
    (b) we just revealed the actual serial number inside of it
  - Revealing the serial number prevents double spending
  - The trick is doing this efficiently!

Spending Zerocoin

- Zero knowledge [Goldwasser, Micali 1980s, and beyond]
  - Prove a statement without revealing <u>any other information</u> (other than that a statement is true)

Spending Zerocoin

- Zero knowledge [Goldwasser, Micali 1980s, and beyond]
  - Prove a statement without revealing <u>any other information</u>
  - Here we prove that:
    (a) there exists a Zerocoin (commitment) in the block chain
    (b) the thing we revealed is the opening to that commitment
  - Revealing the serial number prevents double spending
  - The trick is doing this efficiently!

Spending Zerocoin

- Possible proof statement (not efficient, see CryptoNote):
  - Public values: list of Zerocoin commitments $C_1, C_2, \ldots, C_N$
    Revealed serial number *SN*
  - Prove you know a coin *C* and randomness *r* such that:

$$C = C_1 \ \vee \ C = C_2 \ \vee \ \ldots \ \vee \ C = C_N$$
$$\wedge \ C = Commit(SN; r)$$

  - Problem: using standard techniques, this ZK proof has cost/ size O(*N*)

Spending Zerocoin

- Zerocoin (actual protocol)
  - Use an efficient RSA one-way accumulator
  - Accumulate $C_1, C_2, \ldots, C_N$ to produce a short value A
  - Then prove knowledge of a short witness s.t. $C \in inputs(A)$
  - And prove knowledge that C opens to the serial number

Requires a DDL proof (**~25kb**) for each spend. In the block chain.

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Anonymity set comparison

- Anonymity set in CoinJoin:
  - **M**: where **M** is number of inputs in the transaction (bounded by TX size)
- Anonymity set in ByteCoin/RingCT:
  - **N**: where **N** is the number of inputs allowed in a transaction (bounded by TX size, 7-11 historically)
- Anonymity set in Zerocoin:
  - **P**: where **P** is number of total Zerocoins minted on the blockchain thus far* (independent of TX size)

Next time

- So far we've given a very high level view of Zerocoin, CryptoNote/RingCT
  - Next time we'll talk about Zerocash (Zcash) and MimbleWimble
  - As well as other online techniques like Tumblebit